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Project: An Intelligent Decision Support System for USA Stores

OBJECTIVES:

- Apply Forecasting and Optimization methods in a complex real-world task, managing and documenting the project.
- Corresponds to 75% of the final Course Unit grade;

> Teams

- **Preferably teams / groups of 4 students**.
- **All team members** need to develop the project**.**
- Each team needs to elect a **leader** (responsible for managing the team and assuring a good development of the project through the semester).
- During classes there will be an **assessment** of the team behavior..

> Project Group Self-Assessment (A)

- At the end of the project execution, each team proposes a final project grade, from 0 (0%) to 20 “valores” (100%): **A**. The value of A should be justified. **Example** “The team believes that the project deserves 16 ”valores”, because we executed well these objectives..., but we were unable to achieve the objectives...”

- Each project is evaluated by the teacher, resulting in grade **P**, which can be equal or different than **A**.

Notes:
- Accomplishing all the objectives should result in a good evaluation, but not necessarily an excellent grade (e.g., 19 “valores”). The quality of the achieved results (programing code, analysis, report and presentation) is also relevant.
- It is not necessary to achieve all project objectives to be approved for the Curricular Unit.

> Project Individual Assessment

- Each group can propose an **individual self-differentiation**, clarifying the individual contributions for each member of the group.
- **Individual self-differentiation** follows these rules:
    - the individual sum of the grades cannot be greater than M x P, where M is the number of group members and P is the final grade of the project (assessed by the teacher).
    - a student cannot fail due to an increase of other grades (to do this, the group would have to “dismiss” the student in advance);
- Individual self-differentiation must be **justified in the report**, with a description of individual performance (see slide about report).
- The final decision made by the teacher (for example, via attendance and participation in practical classes, performance in presenting the project).

Example: for a P grade of 14 and 4 students *A, B, C, D*:
- *A* performed better than the average, {*B, C*} contributed with an average performance and D contributed less: A=15, B=14, C=14, D=13.

Ethical rules and concerns

Do **not falsify results**, **nor plagiarize projects from other groups** or **use Internet content without being properly identified and referenced** (who is the author, where it was published). The use of other sources cannot be exaggerated given the total work developed. Students are responsible for ensuring that all work submitted is original and references consulted are properly cited.

Artificial Intelligence (AI) tools, such as Chat-GPT, Gemini and others, **can be used but with caution**. Any copied and pasted knowledge from AI tools should be **understandable** by the group members (e.g., to know what it does). Moreover, the **use of AI tools must be made explicit in the report and code** (e.g., which queries were used by which tools, which results were obtained). This information provides transparency and accountability for the use of AI in academic work and helps to avoid potential plagiarism issues.

> Data

- **Daily values** from 4 USA based physical stores, located in: Baltimore, MD; Lancaster, PA; Philadelphia, PA; and Richmond, VA.
- For each day, the database (one per store) includes:
  - **Date** – day of the year (exogenous)
  - **Num_Employees**– number of employees in store (exogenous).
  - **Num_Customers** – number of daily customers in store (target)
  - **Pct_On_Sale** - percentage of products offered at a reduced price (exogenous).
  - **TouristEvent** - if true, there was a touristic event that could attract customers to store (No/Yes, exogenous)
  - **Sales** – the daily number of product sales (endogenous).

> Data

Available via four CSV files: baltimore.csv, lancaster.csv, philadelphia.csv and richmond.csv.

In R, the data can be loaded using the command:

```
> bt=read.table("baltimore.csv",header=TRUE,sep=",")
> summary(bt)
     Date           Num_Employees    Num_Customers      Pct_On_Sale    TouristEvent
 Length:714         Min.   : 0.00   Min.   :   0.0   Min.   : 0.00   Length:714
 Class :character   1st Qu.:12.00   1st Qu.:  76.0   1st Qu.: 8.62   Class :character
 Mode  :character   Median :16.00   Median : 106.0   Median :10.55   Mode  :character
                    Mean   :17.99   Mean   : 138.0   Mean   :11.25
                    3rd Qu.:21.00   3rd Qu.: 150.8   3rd Qu.:13.06
                    Max.   :83.00   Max.   :1088.0   Max.   :28.01
                                                     NA's   :1
     Sales
 Min.   :     0
 1st Qu.: 31203
 Median : 36647
 Mean   : 40267
 3rd Qu.: 44636
 Max.   :171345
```

> Forecasting Goal:

Forecast the number of daily customers for each store using a maximum horizon event of one week.

- Both univariate and multivariate time series models can be adopted, as well as mathematical and machine learning algorithms.
- Models should be trained on older data and tested on newer data.
- The time horizon must involve a prediction from 1 to 7 days ahead (one week, up to H=7 multi-step ahead predictions).

Forecasting requirements:

- At least, each group member should model (code, experiment, test) one different forecasting method.
- The quality performance measures and evaluation procedures should be defined by the group.

Better assessment if different and more interesting forecasting methods are explored.

> Store management

The management of physical stores includes setting several issues, including:

- Managing inventory and stocks;
- Store layout;
- Customer service;
- Financial oversight;
- …
- **Hiring employees (Human Resources);**
- **Setting** sales and **promotions;**

> Human Resources (HR)

Uses 2 main types of HR:

a) **Junior (J)** – trainees; Daily salary: 60 USD ($) if weekday, 70 USD if weekend; can "help" with a factor of F=$\mathbf{F_J}$; can assist with quality 6 clients.

b) **eXpert (X)** – with at least 6 months of experience: 80 USD if weekday, 95 USD if weekend; can "help" with a factor of F=$\mathbf{F_X}$; can assist with quality 7 clients.

HR are hired for each store s and day d, resulting in a total of: $HR_{s,d}=J_{s,d}+X_{s,d}$ employees (in the CSV files, this is equivalent to *Num_Employees*).

The daily number of **assisted customers** is:

$A_{s,d}= \min(\mathbf{7*X_{s,d}} +\mathbf{6*J_{s,d}}; C_{s,d})$, where $\mathbf{C_{s,d}}$ is the predicted number of customers (*Num_Customers*)

Only **assisted customers** can buy products.

For a given d, there are: **c** in $\{1,\ldots,A_{s,d}\}$ assisted customers.

When computing daily assisted customers, first all **X workers are used**, then if needed, the remainder **J workers** are used.

> Human Resources (HR)

The “help” factor changes for each store (some stores are more “modern” and have better “managers”):

- Baltimore: F_j=1.00, F_x=1.15, fixed week cost W_s=700;
- Lancaster: F_j=1.05, F_x=1.20, fixed week cost W_s=730;
- Philadelphia: F_j=1.10, F_x=1.15, fixed week cost W_s=760;
- Richmond: F_j=1.15, F_x=1.25, fixed week cost W_s=800.

> Promotions

For each store s and day d, it is necessary to define the **PR**omotion percentage (from 0% to 30%) of articles that are offered at a reduced price $\mathbf{PR_{s,d}}$ (in the CSV files, this is equivalent to *Pct_On_Sale*).

> Sold units and profit

For each store s and day d, the number of sold units per assisted customer **c** is:

$\mathbf{U_{s,d,c}}$= **round**(F*10/ln(2-$PR_{s,d}$)), where ln denotes the natural logarithm and F is the HR factor (F_J or F_X).

The **sold unit profit** ($) for an assisted customer **c** is:

$\mathbf{P_{s,d,c}}$= **round**($U_{s,d,c}$ * (1-$PR_{s,d}$) * 1.07)

The **daily store return profit** is given by:

$\mathbf{R_{s,d}} = \text{sum\_i} \ (P_{s,d,i}) - \boxed{J_{s,d}*\text{hr\_}cost - X_{s,d}*\text{hr\_}cost}$ *HR costs*

The final **weekly store profit** ($) is given by:

$\mathbf{R_s} = \text{sum\_d} \ (R_{s,d}) - \boxed{W_s}$ *Fixed store costs.*

> Optimization Goals:

At the end of day d, the aim is to define a **weekly plan** for all stores (HW and promotions) for the next 7 days: between days d+1 and d+7. The resource plan needs to include the daily integer numbers of J and X resources, the percentage of promoted products (PR) for each store.

There are 3 main objectives:

O1 - **Maximize** the **total return profit (R_s)** for the planned week**.**

O2 - **Maximize O1** assuming a **hard constraint: maximum of 10,000 sold units** for **all four stores!**

O3 – **Maximize O2 and Minimize the total number of HR resources.**

Store plan example for baltimore

```
baltimore :
          Date Num_Customers
708 2014-06-08            97
709 2014-06-09            61
710 2014-06-10            65
711 2014-06-11            71
712 2014-06-12            65
713 2014-06-13            89
714 2014-06-14           125
```

Weekend (2014-06-08) — Weekend (2014-06-14)

```
>> Weekly plan:
day:     su     mo     tu     we     th     fr     sa
PR:      0.00   0.05   0.10   0.15   0.20   0.25   0.30
HR X:    4      0      8      20     0      4      3      tot.=   39
HR J:    0      10     4      0      5      5      4      tot.=   28
weekly HR                                                 tot.=   67
[1] "Assisted customers:"
 by X:   28     0      56     71     0      28     21     tot.=   204
 by J:   0      60     9      0      30     30     24     tot.=   153
[1] "Computed estimates:"
Xunits:  476    0      1008   1349   0      588    462    tot.=   3883
Junits:  0      900    144    0      510    540    456    tot.=   2550
Xsales:  509    0      971    1227   0      472    346    tot.=   3525
Jsales:  0      915    139    0      437    433    342    tot.=   2266
XHRc:    380    0      640    1600   0      320    285    tot.=   3225
JHRc:    0      600    240    0      300    300    280    tot.=   1720
>> week profit= 3525 + 2266 - 3225 - 1720 - 700 = 146
```

Elements of the plan

Impacts O3

Impacts O2,O3

Impacts O1,O2,O3

Store plan example for philadelphia

```
philadelphia :
          Date Num_Customers
708 2014-06-08           230
709 2014-06-09           144
710 2014-06-10           154
711 2014-06-11           168
712 2014-06-12           154
713 2014-06-13           211
714 2014-06-14           298
>> Weekly plan:
day:    su     mo     tu     we     th     fr     sa
PR:     0.00   0.05   0.10   0.15   0.20   0.25   0.30
HR X:   4      0      8      20     0      4      3      tot.=   39
HR J:   0      10     4      0      5      5      4      tot.=   28
weekly HR                                                tot.=   67
[1] "Assisted customers:"
 by X:  28     0      56     140    0      28     21     tot.=   273
 by J:  0      60     24     0      30     30     24     tot.=   168
[1] "Computed estimates:"
Xunits: 476    0      1008   2660   0      588    462    tot.=   5194
Junits: 0      960    408    0      570    600    504    tot.=   3042
Xsales: 509    0      971    2419   0      472    346    tot.=   4717
Jsales: 0      976    393    0      488    482    377    tot.=   2716
XHRc:   380    0      640    1600   0      320    285    tot.=   3225
JHRc:   0      600    240    0      300    300    280    tot.=   1720
>> week profit= 4717 + 2716 - 3225 - 1720 - 760 = 1728
```

Weekend (708 2014-06-08)

Weekend (714 2014-06-14)

Elements of the plan

Impacts O3

Impacts O2,O3

Impacts O1,O2,O3

Two important notes:

1. The two previous slides (#15 and #16) were provided to improve debug issues: to make sure the profit and HR resource functions are well computed. They assume the exact same plan for 2 different stores. In practice, you should obtain different optimized plans for different stores.
2. While O1 can be solved independently for each store, O2 and O3 cannot. Thus, it saves computing code if you also assume the same set of parameters to be optimized for all three goals (O1, O2 and O3).

Optimization requirements:

- For the optimization, assume:
  - forecasted values for the next 7 days (**best**) OR
  - use actual values;
- Assume up to 3 optimization possibilities (O1, O2 and O3).
- At least one optimization method must be used by each group member. Each member must be responsible for implementing (code), configuring (setting parameters) and experimenting with this optimization method.

What is evaluated?

- Quality of the forecasting and optimization study (**report**).
- System developed, including a console or graphical (better) mode interface (e.g., via shiny.rstudio.com) to demonstrate the complete Intelligent Data Analysis system (prediction and optimization) at work.
- Better if:
  - several forecasting / optimization methods are compared;
  - several R packages are explored;
  - forecasts are used by the optimization methods;
  - a more robust evaluation is adopted;
  - more realism is introduced (e.g., other optimization scenarios);
  - study the effect of changing some of the forecasting / optimization parameters.

> Project Execution

- From today until the date of the project presentation.
- Teacher Support is performed during AID classes.
- During the classes, a preliminary evaluation (non-formal, with a reduced weight) will be carried out for each group and project (all members of the group should be present): this contributes to the **PA*** project assessment component.

* PA - Active Participation assessment.

> Week Summary:

From 2026-03-16, all groups must show the teacher a “weekly project work summary report” (preferably via Googledocs, sharing a link with the option “anyone with the link can view”).

The summary should include 2 parts:

- Work team summary: what was accomplished, how many group meetings were held and their duration (in hours), examples of outputs (e.g., code, graphs, tables) and list of doubts.
- For each element of the project: what you did individually and how many hours you dedicated to the project during the week.

> Week Summary example:

Semana 4 (11/11/2025 - 17/11/2025)

Trabalho de Grupo	
Resumo do Trabalho Realizado: Ao longo da semana, o grupo concentrou-se em [redacted]	
Número de Reuniões Realizadas: 3	
Reunião Nº1	**Dia:** 11/11/2025
	Duração: 1h 30 min
	Objetivos: - Discussão das sugestões dadas pelo professor [redacted]
Reunião Nº2	**Dia:** 15/11/2025
	Duração: 1h
	Objetivos: - Discussão e validação do trabalho realizado individualmente

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Dúvida	Esclarecida?
Parâmetros do [redacted]?	☑
Faz sentido não ter [redacted] antes de [redacted]?	☑

Próximos Passos:

- Implementação do melhor modelo [redacted]

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Trabalho Individual		
Elemento	**Tarefas Realizadas**	**Horas dedicadas ao projeto**
A[redacted] PG[redacted]	- Implementação das sugestões dadas pelo professor e aperfeiçoamento dos modelos de previsão adotados [redacted]	5h
Ana [redacted] PG[redacted]	[redacted] - Escolha do melhor modelo de entre as abordagens testadas, [redacted]	5h

Resultados Obtidos:

[redacted]

Modelo <chr>	Tipo <chr>	k <dbl>	Media_NMAE <dbl>	Mediana_NMAE <dbl>
[redacted]				

. . .

> Project upload and presentation

- **Dates:** 2025-05-25 (**MECD**) or 2026-05-27 (**MEGSI**)
- **Group presentation:** in-class with all elements, includes project presentation and question answering. Only one group will be present at the classroom for each presentation. A group schedule will be provided at the elearning system.
- **Project upload:** Report (in pdf) – mandatory; presentation slides (in pdf format) – optional; R code attached to the zip file - mandatory => single .zip file to be uploaded in the elearning system.

Demo Video

- A computer video must be recorded (maximum of 5 minutes, ideally with voice narration) where the developed system is demonstrated (in R and/or other tools) and uploaded to **youtube**.
- **Include in the report** the link of the youtube video.

To search for free screen video capturing software do a google search:

screen capture video

> Report

- **Cover**

1. **Introduction:** brief description of the group, project goals and how the report is structured.
2. **Project execution**: project planning, how the group worked, self-assessment (**A** grade proposed for the project), proposal of individual self-assessment, etc. Use half a page per group element, describing what she/he did in terms of the project execution (e.g., effort, tasks, code, tests).
3. **Prediction task**: what was done and obtained results.
4. **Optimization task**: what was done and obtained results.
5. **System demonstration**: of the developed intelligent data analysis system, use 1 sentence with the youtube link.
6. **Conclusions**: brief appreciation and discussion of what has been accomplished

- **Bibliography**: optional but desired
- **Attachments**: optional, can include further graphs, tables, examples of AI prompts and results.

Try to have a maximum of around 20 to 40 pages of the body document (without cover, indexes, bibliography and attachments). Use a direct and concise writing.